10.18

(a)

```
> cat("Percentage of mothers with some college:", pct_mother, "%\n")
Percentage of mothers with some college: 12.14953 %
> cat("Percentage of fathers with some college:", pct_father, "%\n")
Percentage of fathers with some college: 11.68224 %
```

(b)

```
                educ mothercoll fathercoll
educ       1.0000000  0.3594705  0.3984962
mothercoll 0.3594705  1.0000000  0.3545709
fathercoll 0.3984962  0.3545709  1.0000000

                 educ motheredu fathereduc
educ        1.0000000  0.3870198  0.4154030
mothereduc  0.3870198  1.0000000  0.5540632
fathereduc  0.4154030  0.5540632  1.0000000
```

cov(educ, mothercoll) = 0.3595 , cov(educ, fathercoll) = 0.3985

⇒ positive correlations suggest that the instruments are relevant.

Mothercoll, fathercoll might be better instruments than mothereduc and fathereduc.

因为跨入大學是一個門檻,更能反映對子女升學的重視程度.

(c) $\hat{\beta}_{educ} = 0.0760$

95% CI = (-0.0012, 0.1535)

```
          2.5 %    97.5 %
educ -0.001219763 0.1532557
```

(d)

```
Analysis of Variance Table

Model 1: educ ~ exper + I(exper^2)
Model 2: educ ~ mothercoll + exper + I(exper^2)
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    425 2219.2
2    424 1929.9  1    289.32 63.563 1.455e-14 ***
```

F-test = 63.563 > 10 (rule-of-thumb threshold)

⇒ mothercoll is a strong instrument.

(e) $\hat{\beta}_{educ} = 0.0878$

95% CI = (0.0275, 0.1482)

This interval is narrower than one in part (c).

```
          2.5 %    97.5 %
educ 0.02751845 0.1481769
```

(f)

```
Analysis of Variance Table

Model 1: educ ~ exper + I(exper^2)
Model 2: educ ~ mothercoll + fathercoll + exper + I(exper^2)
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    425 2219.2
2    423 1748.3  2    470.88 56.963 < 2.2e-16 ***
```

H_0 :

F-test = 59.963 > 10

⇒ The instruments seem adequately strong.

(g)

```
Diagnostic tests:
                 df1 df2 statistic p-value
Weak instruments   2 423    56.963  <2e-16 ***
Wu-Hausman         1 423     0.519   0.472
Sargan             1  NA     0.238   0.626
```

$NR^2 = 0.238$, p-value = 0.626 > 0.05

We do not reject H_0 that the instruments are valid. It is no evidence that the surplus infrument is invalid.

10.20

(a) $\hat{\beta} = 1.2018 > 1$

```
(Intercept)           x
0.003249601 1.201839753
```

$\Rightarrow$ Microsoft is riskier than the market portfolio.

(b)

```
Call:
lm(formula = x ~ rank, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.110497 -0.006308  0.001497  0.009433  0.029513 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -7.903e-02  2.195e-03   -36.0   <2e-16 ***
rank         9.067e-04  2.104e-05    43.1   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom
Multiple R-squared:  0.9126,	Adjusted R-squared:  0.9121 
F-statistic:  1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

First stage: $\widehat{MKT-RF} = -0.07903 + 0.0009067 RANK$

p-value < 0.01, $R^2 = 0.9126$, F-test $= 1858$

$\Rightarrow$ Rank is a strong IV.

(c)

```
Call:
lm(formula = y ~ x + vhat, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27140 -0.04213 -0.00911  0.03423  0.34887 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.003018   0.005984   0.504   0.6146    
x            1.278318   0.126749  10.085   <2e-16 ***
vhat        -0.874599   0.428626  -2.040   0.0428 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,	Adjusted R-squared:   0.36 
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

$\hat{\gamma} = -0.8746$, p-value $= 0.0428$

It is significant at 5% but not at 1%. Thus, we can conclude that the market return is exogenous.

(d)

```
Call:
ivreg(formula = y ~ x | rank, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.271625 -0.049675 -0.009693  0.037683  0.355579 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003018   0.006044   0.499    0.618    
x           1.278318   0.128011   9.986   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared: 0.3508,	Adjusted R-squared: 0.3472 
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16
```

$\hat{\beta}_{IV} = 1.2783 > \hat{\beta}_{OLS} = 1.2018$. This agrees with the expectation that OLS is biased downward due to measurement error.

(e)

```
Call:
lm(formula = x ~ rank + pos, data = capm5)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.109182 -0.006732  0.002858  0.008936  0.026652 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -0.0804216  0.0022622  -35.55   <2e-16 ***
rank         0.0009819  0.0000400   24.55   <2e-16 ***
pos         -0.0092762  0.0042156   -2.20   0.0291 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,	Adjusted R-squared:  0.9139 
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

$R^2 = 0.9149$, $F = 951.3$, p-value < 0.01.

We can conclude that we have adequately strong IVs.

(f)

```
Call:
lm(formula = y ~ x + vhat2, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27132 -0.04261 -0.00812  0.03343  0.34867 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.003004   0.005972   0.503   0.6157    
x            1.283118   0.126344  10.156   <2e-16 ***
vhat2       -0.954918   0.433062  -2.205   0.0287 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,	Adjusted R-squared:  0.3625 
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

p-value = 0.0287 > 0.01

We do not reject $H_0: \delta = 0$ that it is exogeneity.

(g)

```
Call:
ivreg(formula = y ~ x | rank + pos, data = capm5)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.27168 -0.04960 -0.00983  0.03762  0.35543 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.003004   0.006044   0.497     0.62    
x           1.283118   0.127866  10.035   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom
Multiple R-Squared: 0.3507,	Adjusted R-squared: 0.347 
Wald test: 100.7 on 1 and 178 DF,  p-value: < 2.2e-16 
```

$\hat{\beta}_{IV} = 1.2831 > \hat{\beta}_{OLS} = 1.2018$

The measurement error was present and the OLS was underestimate β.

(h) $NR^2 = 0.5585$

p-value = 0.4549 > 0.05

We do not reject the null that the surplus IV is invalid.

```
> NR2
[1] 0.5584634
```

```
> p_value
[1] 0.45488
```